Osmosis and Osmotic pressure of the solution

21. Osmotic pressure of a solution can be measured quickly and accurately by
(a) Berkeley and Hartley's method
(b) Morse's method
(c) Pfeffer's method
(d) De Vries method

22. The solution in which the blood cells retain their normal form are with regard to the blood
(a) Isotonic
(b) Isomotic
(c) Hypertonic
(d) Equinormal

23. The osmotic pressure of a solution is given by the relation
(a) $P = \frac{RT}{C}$ (b) $P = \frac{CT}{R}$
(c) $P = \frac{RC}{T}$ (d) $\frac{P}{C} = RT$

24. The osmotic pressure of a solution is directly proportional to
(a) The molecular concentration of solute
(b) The absolute temperature at a given concentration
(c) The lowering of vapour pressure
(d) All of the above

25. What would happen if a thin slice of sugar beet is placed in a concentrated solution of $NaCl$
(a) Sugar beet will lose water from its cells
(b) Sugar beet will absorb water from solution
(c) Sugar beet will neither absorb nor lose water
(d) Sugar beet will dissolve in solution

26. The osmotic pressure of a dilute solution is given by
(a) $P = P_o x$ (b) $\pi V = nRT$
(c) $\Delta P = P_o N_2$ (d) $\frac{\Delta P}{P_o} = \frac{P_o - P}{P_o}$

27. Which statement is wrong regarding osmotic pressure (P), volume (V) and temperature (T)
(a) $P \propto \frac{1}{V}$ if T is constant
(b) $P \propto T$ if V is constant
(c) $P \propto V$ if T is constant
(d) PV is constant if T is constant

28. Isotonic solutions have
(a) Equal temperature
(b) Equal osmotic pressure
(c) Equal volume
(d) Equal amount of solute

29. Which of the following associated with isotonic solutions is not correct
(a) They will have the same osmotic pressure
(b) They have the same weight concentrations
(c) Osmosis does not take place when the two solutions are separated by a semipermeable membrane
(d) They will have the same vapour pressure

30. Isotonic solution have the same
(a) Density
(b) Molar concentration
(c) Normality
(d) None of these

31. A 0.6% solution of urea (molecular weight = 60) would be isotonic with
(a) $0.1M$ glucose
(b) $0.1 MKCl$
(c) 0.6% glucose solution
(d) $0.6\% KCl$ solution

32. The value of osmotic pressure of a 0.2 *M* aqueous solution at 293*K* is
(a) 8.4 *atm* (b) 0.48*atm*
(c) 4.8 *atm* (d) 4.0 *atm*

33. Diffusion of solvent through a semi permeable membrane is called
(a) Diffusion
(b) Osmosis
(c) Active absorption
(d) Plasmolysis

34. Solutions having the same osmotic pressure under a given set of conditions are known as
(a) Hypertonic (b) Hypotonic
(c) Normal (d) Isotonic

35. At low concentrations, the statement that equimolal solutions under a given set of experimental conditions have equal osmotic pressure is true for
(a) All solutions
(b) Solutions of non-electrolytes only
(c) Solutions of electrolytes only
(d) None of these

36. Which one of the following would lose weight on exposure to atmosphere
(a) Concentrated H_2SO_4
(b) Solid $NaOH$
(c) A saturated solution of CO_2
(d) Anhydrous sodium carbonate

37. The molecular weight of $NaCl$ determined by osmotic pressure method will be
(a) Same as theoritical value

(b) Higher than theoritical value
(c) Lower than theoritical value
(d) None of these

38. The osmotic pressure of solution increases, if
(a) Temperature is decreased
(b) Solution concentration is increased
(c) Number of solute molecules is increased
(d) Volume is increased

39. At the same temperature, following solution will be isotonic
(a) 3.24 *gm* of sucrose per litre of water and 0.18 *gm* glucose per litre of water
(b) 3.42 *gm* of sucrose per litre and 0.18 *gm* glucose in 0.1 litre water
(c) 3.24 *gm* of sucrose per litre of water and 0.585 *gm* of sodium chloride per litre of water
(d) 3.42 *gm* of sucrose per litre of water and 1.17 *gm* of sodium chloride per litre of water

40. The osmotic pressure of a decinormal solution of $BaCl_2$ in water is
(a) Inversely proportional to its celsius temperature
(b) Inversely proportional to its absolute temperature
(c) Directly proportional to its celsius temperature
(d) Directly proportional to its absolute temperature